Problem Chosen	2026 HiMCM/MidMCM Summary Sheet	Team Control Number
B		18588

CP-SET: Measuring Useful Cooling in Open Public Spaces

Summary

An open cooling system should be judged by the useful service reaching its users. We develop the **Coupled Psychrometric-SET Service Efficiency & Effectiveness Framework (CP-SET)** to connect heat and moisture balances, local air delivery, human thermal response and operational decisions.

The freezer case establishes four essential quantities: service quantity, local quality, duration and electrical input. Its useful storage service motivates a human-service analogue: signed SET-excess relief accumulated over passenger-hours, divided by the electricity used by every operating device. This distinguishes efficiency from the adequacy of the cooling delivered.

For a New York City bus-shelter geometry, the framework combines wet-bulb-limited evaporation, dry-air-weighted local mixing and Standard Effective Temperature. At 34°C, fan A achieves efficiencies of 84.58 and 90.57 person·°C·h/kWh at 30% and 70% RH. B and C require 4.8 and 10 times A's accumulated relief to overcome their power ratios. A delivery-loss test identifies where B overtakes A. A separate hot-dry test shows why larger thermal improvement need not imply higher efficiency.

At Canberra MRT station, the same delivery and response modules are retained while zone enthalpy, humidity and surface temperature evolve with exchange, passenger loads and equipment operation. Across 24 sampled installations, A3 reaches the highest efficiency, 37.58; among six equal-coverage benchmarks, A36 leads at 35.49. Relocating the same AB30 inventory toward passenger demand raises its efficiency from 15.39 to 18.39. Coverage and local delivery therefore matter alongside the rated cooling effect.

At **Fremont Street Experience** and the **Honolulu Zoo ticket queue**, modeled service areas differ by a factor of 139.7. Largest-fitting pure A/B/C plans achieve efficiencies of 11.88/5.06/5.72 and 47.71/21.28/20.63, respectively. A controlled climate exchange confirms the B–C reversal. An energy-weighted bound establishes each site's maximum efficiency, while scale and boundary tests separate device replication from lost coverage and clipped delivery.

Extending the same physical outputs to thermal performance, visitor access, annual energy and cost gives an integer allocation problem. The stated operator policy selects A266C16 in Fremont and A1B1 in Honolulu, at equivalent annual costs of USD 50,926 and 627. In Honolulu, relaxing required mean relief from 1.50 to 1.49°C selects three fans instead: the cost-minimizing choice depends on the service threshold.

Conservation checks, limiting-case recovery and time-step refinement verify the numerical implementation. Climate, local delivery and cost sensitivities identify the assumptions that affect the decision. The resulting memorandum and public explanation translate useful cooling per kWh into equipment placement, operating schedules and extreme-heat procedures.

Keywords: psychrometrics; local air delivery; Standard Effective Temperature; cooling service efficiency; multi-objective decisions.

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1 Introduction

1.1 Background

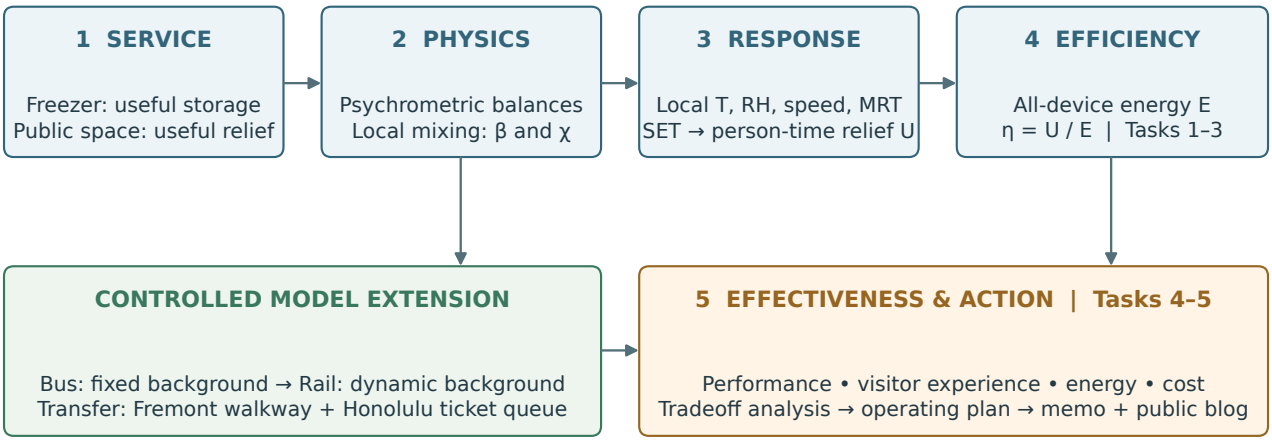
Open cooling systems must deliver relief despite continuous exchange with their surroundings. Fans alter human convective and evaporative losses; evaporative coolers also change air temperature and moisture [6, 5]. Comparing them requires useful service per unit energy.

1.2 Problem Restatement

The five tasks are to define cooling efficiency, develop bus and rail models, test two contrasting venues, extend efficiency into an effectiveness decision, and communicate operating guidance through a memorandum and public blog [4].

1.3 Our Modeling Framework

Our **CP-SET Framework (Coupled Psychrometric-SET Service Efficiency & Effectiveness Framework)** connects conserved heat and moisture to local exposure, then converts SET relief into person-time service per kWh. Requirements 1–3 compare efficiency; Requirement 4 adds cooling adequacy, visitor experience and cost. The workflow below preserves this progression.



CP-SET workflow. A shared physical and service model supports progressively broader decisions.

AI assistance in model development, coding and writing is documented in the appended report [25, 26].

2 Assumptions and Justifications

A1. Matched service boundary. Comparisons share weather, people, geometry and period; the problem's kW/hr entries are interpreted as kW. *Justification:* A common workload and complete device-energy accounting isolate strategy efficiency.

A2. Zoned air and rapid local mixing. Zones have uniform background states; local supply mixes at negligible storage, with disjoint footprints and separately prescribed speed. *Justification:* This resolves occupied-location differences when local residence time is short relative to the 120-s station disturbance.

A3. Conserved heat and water. Evaporation is near-adiabatic; background dry-air mass is fixed and exchange is balanced; shaded radiation uses mean radiant temperature. Condensation is excluded in public-space cases. *Justification:* These approximations retain the dominant transfers while counting every electrical and vapor-enthalpy source once.

A4. Reference human response. SET uses standing transport users and venue visitors at 1.2 met and 0.5 clo, with a 26°C excess reference. *Justification:* One signed-relief measure compares airflow and evaporation. Dry-heat tests use $T_a = T_r = 39/40$°C and RH 15/20%: 40°C is the implementation boundary; the interior 39°C test retains the ranking [8].

A5. Explicit design inputs. Published sources establish site geometry and climate context; tabulated estimates specify operating and economic conditions. *Justification:* Reproducible inputs support sensitivity analysis. Costs use one currency and horizon; equipment clearance is the visitor-access proxy. SET relief and operational heat-safety procedures serve separate purposes.

Table 1: Transport-case inputs. Geometry sources are cited; remaining entries are model design assumptions.

Input	Bus shelter	Canberra waiting area
Geometry	4.2672×1.5240×2.7178 m [10]; three equal gross strips	L = 220 m [17]; effective waiting width 6 m, usable fraction .8, air height 4 m; three zones
Climate / radiation	T_a = 34°C; RH 30/70%; T_r = 36°C	T_o = 33°C, RH 70%; external $T_{r,o}$ = 33°C [18]
People / time	$(4, 4, 4)$ people, 1 h; redistribution $(8, 3, 1)$ or $(1, 3, 8)$	$(26.4, 44, 17.6) \rightarrow (88, 198, 66)$ linearly per 15-min cycle; 2-h warm-up + 2-h evaluation
Exchange	Fixed outdoor background	ACH $(8, 4, 10)$ plus $(8, 4, 12)$ in last 2 min; adjacent flow .88 kg dry air/s
Surface / people	Prescribed radiant conditions	Per zone C_m = 17.6 MJ/K; H_{ma}/H_{mo} = 880/1760 W/K; absorbed load 3520 W; F = (.55, .85, .55); per person q_c/q_r = 45/30 W, $g = 45/(2.45 \times 10^6)$ kg/s
Device delivery	Profiles in Table 2; $\chi = 0$; χ sensitivity uses A β = (.8, .45, .1) and rail flows	β = .5, χ = 1; A/B/C flow .6/.6/1.2 kg dry air/s; footprints 20/30/40 m^2
Shared response	Pressure 101325 Pa; v_0 = .2 m/s; body area 1.8258 m^2, external work zero; $\varepsilon_{B/C}$ = .7; outlet drops capped at 5/8°C. Θ_c sensitivity 24/26/28°C.	

Table 2: Local speed and mixing profiles in zone order; each entry describes an independent local state.

Device	Bus speed (m/s)	Bus $\beta_{s,z}$	Rail covered speed (m/s)
A	1/.55/.25	Immaterial at $\chi = 0$	.60/1/.50
B	.80/.50/.30	.80/.45/.10	.50/.80/.45
C	1/.80/.50	.90/.70/.40	.75/1/.60

Table 3: Venue design inputs. The canopy length and site identities are sourced; occupied width, queue area, weather snapshots, attendance, delivery and prices are planning assumptions.

Input	Fremont Street Experience	Honolulu Zoo ticketing queue
Service boundary	419.1-m canopy [19]; assumed 20-m occupied width; $A_z = (2400, 3600, 2382)$ m^2	Window-side shaded public queue [20, 21]; one assumed 10×6 m demand area
Climate / radiation	$T_a/T_r = 38/39$°C; RH 20%; $v_0 = .20$ m/s [22]	Warm, weak-wind case: $31/33$°C; RH 65%; $v_0 = .35$ m/s [23]
Hourly occupancy	$(120, 240, 90)$; $(180, 480, 150)$; $(240, 720, 180)$; $(180, 480, 150)$	12, 24, 36, 24 simultaneous visitors over four successive one-hour intervals
Common physics	Prescribed outdoor background; $\chi = 0$; standing visitors, 1.2 met, .5 clo. Each complete device footprint has three equal-area delivery strips with Table 2's bus β values and $v = v_0 + (v_{\text{bus}} - .20)$ m/s. Uniform density within each demand zone; disjoint footprints.	
Unit A/B/C	Footprints $20/30/40$ m^2; flow $1.2/2/3$ kg$_{da}$/s; clearance $.5/1/1.5$ m^2; capital USD $300/1000/2000$; annual maintenance USD $30/100/200$.	
Operating basis	$H = 4$ h/session; $H_y = 600$ h/y; equipment life 5 y, discount 5%, zero salvage; USD .20/kWh and 3/m^3; make-up-water factor 1.15.	
Task 4 policy	$q \geq .80$; mean SET-excess relief ≥ 1.5°C; clearance fraction $X \leq .035$; annual energy $\leq 30 \sum_z A_z$ kWh. These conditions enter only the effectiveness decision.	
Transfer tests	Exchange the complete weather tuple at fixed geometry/attendance; scale area and attendance together; clip a single device's far-to-near strips only in the separate boundary test. RH, radiant temperature, natural speed and area are varied individually.	

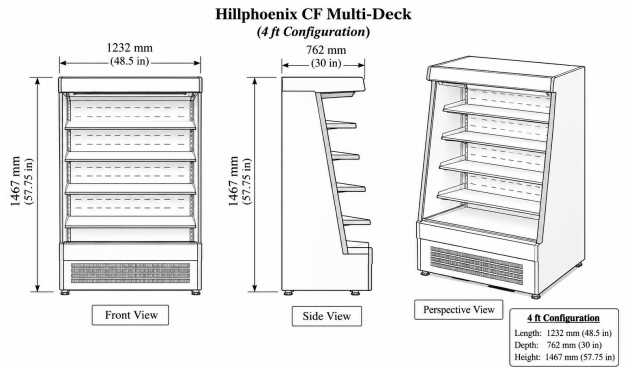
3 Nomenclature

Symbol	Meaning	Unit	Symbol	Meaning	Unit
h	Moist-air enthalpy	J/kg$_{da}$	w	Humidity ratio	kg/kg$_{da}$
$\beta_{s,z}$	Supply dry-air mixing fraction	1	χ	Pre-delivery electrical heat fraction	1
T, T_r	Air / mean radiant temperature	°C	ϕ	Relative humidity	1
Θ, Θ_c	SET / service reference	°C	D	Excess SET above Θ_c	°C
U_s	Accumulated useful relief	person·°C·h	R_s	Accumulated residual excess	person·°C·h
E_s	Electrical energy	kWh	η_s	Cooling service efficiency	person·°C·h/kWh
c_{zk}	Direct-coverage fraction	1	x_{zk}	Device count by zone/type	unit
n_z	Mean simultaneous occupants	person	$\dot{m}_k$	Dry-air supply flow	kg$_{da}$/s
M_z	Zone dry-air mass	kg$_{da}$	p_k, P_k	Device electric power	kW, W
$\overline{R}_s$	Mean residual SET excess	°C	$X_s, C_{s,y}$	Access footprint / annual cost	1, USD/year

Subscripts: s = candidate scheme, z = zone, k = device type (time interval in discrete bus sums), 0 = matched no-cooling baseline, b = background, ℓ = local exposure. Food service uses kg·h/kWh; human service uses the units above.

4 Understanding Cooling Efficiency

An open freezer succeeds by preserving food, not simply by removing heat. Low electricity use can accompany inadequate storage, while large cooling loads may reflect infiltration through the opening. We therefore evaluate *useful service per unit energy*.



The open front admits warm, moist air; cabinet surfaces conduct and radiate heat. Lights, fans, restocking and defrost add operating loads. Insulation and equipment condition modify these mechanisms; equipment age is not a separate score [1]. Product temperatures must be checked across shelves, since a mean can conceal a warm location [2].

Figure 1: Open-case heat pathways and product locations. The supplied medium-temperature display schematic illustrates geometry; freezer limits remain product-specific [11].

The numerator needs **quantity, local quality and duration**: compliant product mass integrated over storage time. The denominator includes refrigeration and auxiliaries without overlapping meter readings. Matched loading, ambient conditions and complete operating cycles prevent short operation or stored cold from inflating efficiency:

$$\boxed{\text{Cooling efficiency} = \frac{\text{useful service accumulated over objects and time}}{\text{energy input over the same period}}.} \tag{1}$$

$$\eta_f = \frac{\int \sum_i m_i(t)\mathbf{1}\{T_{p,i}(t) \in \mathcal{T}_i\}\, dt}{\int p_f(t)\, dt}, \tag{2}$$

Here m_i is product mass (kg), $\mathcal{T}_i$ its admissible temperature range, t is hours and p_f is kW; thus η_f is compliant kg·h/kWh. For people, the response changes to thermal-discomfort relief weighted by occupied time. These recipient-specific numerators cannot be directly compared numerically.

4.1 A shared physical structure

For dry-air mass M_z (kg), balanced dry-air streams $\dot{m}_{jz}$ (kg/s), enthalpy h (J/kg dry air) and humidity ratio w (kg vapor/kg dry air),

$$M_z \dot{h}_z = \sum_j \dot{m}_{jz}(h_j - h_z) + S_{h,z}, \tag{3}$$

$$M_z \dot{w}_z = \sum_j \dot{m}_{jz}(w_j - w_z) + S_{w,z}. \tag{4}$$

S_h (W) and S_w (kg/s) contain the appropriate device, occupant, product and surface sources. Freezer refrigeration can remove heat and condensate; public cooling changes airflow and moisture. The bus fixes the background and solves local delivery; the station retains that calculation while resolving background and surface storage [12]. Figure 2 shows this progression. Cruz-Salas et al. motivate separating treatment, transport and occupied-space conditions [13].

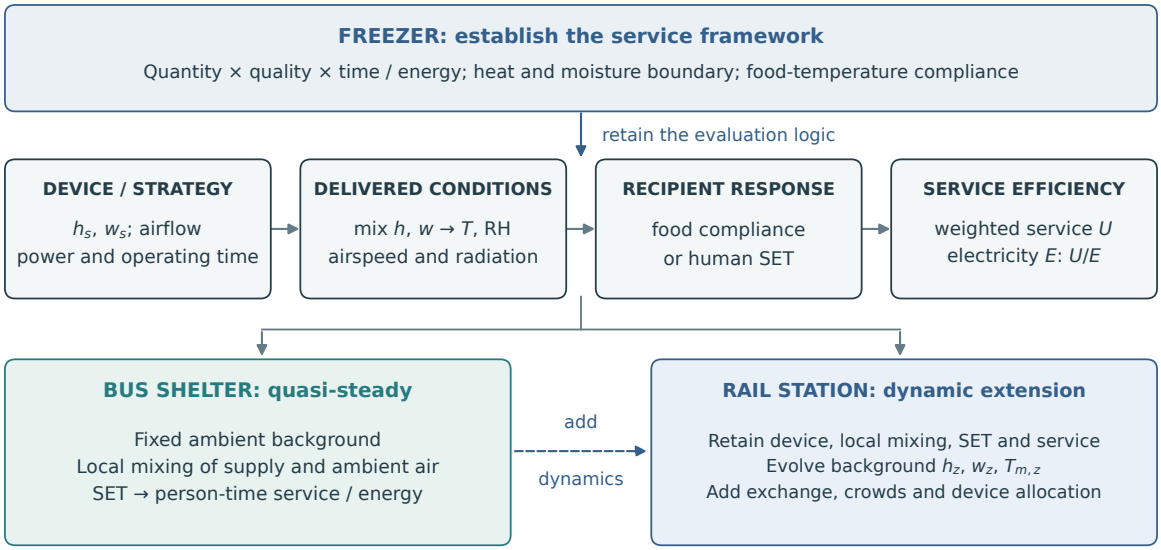


Figure 2: Shared conservation and service accounting, with recipient-specific response and increasingly resolved background dynamics.

5 Bus Stop: Local Delivery and Cooling Efficiency

5.1 Geometry and comparison inputs

NYC DOT's regular shelter measures 14 ft by 5 ft by 8 ft 11 in, or $4.2672 \times 1.5240 \times 2.7178$ m [10]. Its gross footprint is $A = 6.5032\,\mathrm{m}^2$. Three equal strips have $A_z = 2.1677\,\mathrm{m}^2$ and centers $(x_z, W/2)$, $x_z = (z - \frac{1}{2})L/3$; delivery originates at $x = 0$.

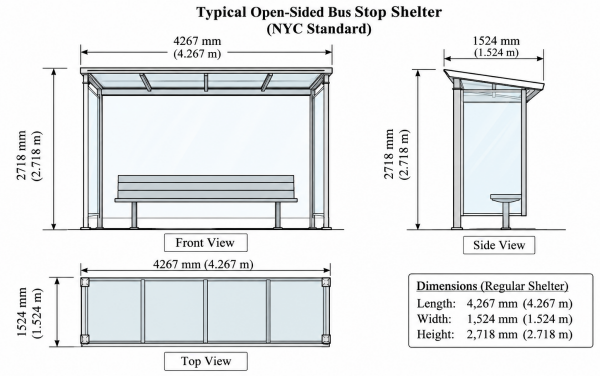


Figure 3: Adopted NYC shelter dimensions [10].

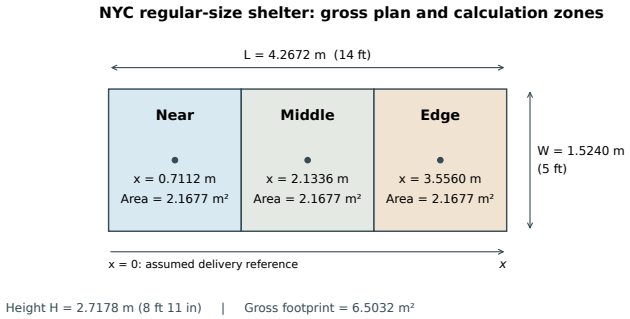


Figure 4: Equal near/middle/edge accounting strips.

The problem's 3 °C fan effect describes perceived cooling; B/C's 5/8 °C describe air cooling [4]. We interpret the listed kW/hr numbers as powers $p_{A/B/C} = 0.25/1.2/2.5$ kW. Tables 1–2 specify the comparison: 34 °C, RH 30/70%, $T_r = 36$ °C and 12 mean occupants. Geometry enters once through $n_z = \rho_z A_z$ when density replaces headcount.

5.2 From device outlet to occupied air

A changes prescribed local speed while retaining background temperature and moisture in the zero-local-heat base case [6]. For B/C, wet-bulb temperature bounds evaporation [5]:

$$T_{o,s} = T_b - \min\{\Delta T_s^{\max},\ \varepsilon_s(T_b - T_{wb,b})\}, \qquad s \in \{B, C\}, \tag{5}$$

b denotes the inlet background; ε_s is evaporative effectiveness and $\Delta T_{B/C}^{\max} = 5/8\,^\circ\mathrm{C}$ are the adopted outlet caps. With ϕ as RH fraction and pressures in Pa, SI psychrometrics give [9]

$$w = 0.621945\frac{p_v}{p - p_v}, \qquad p_v = \phi p_{sat}(T), \tag{6}$$

$$h(T, w) = 1006T + w(2501000 + 1860T), \tag{7}$$

Temperatures are in °C. The near-isenthalpic evaporative stage, $h_{o,s} \simeq h_b$, yields

$$w_{o,s} = \frac{h_b - 1006T_{o,s}}{2501000 + 1860T_{o,s}}. \tag{8}$$

This stage precedes electrical heating and neglects make-up-water enthalpy [3]. A device supply mixes with entrained background in a negligible-storage local region. For $x \in \{h, w\}$,

$$0 = \dot{m}_{s,z}(x_s^{\mathrm{sup}} - x_{\ell,s,z}) + \dot{m}_{e,z}(x_b - x_{\ell,s,z}), \qquad \beta_{s,z} = \frac{\dot{m}_{s,z}}{\dot{m}_{s,z} + \dot{m}_{e,z}}. \tag{9}$$

β is a dry-air flow fraction, not a fraction of passengers or total flow assigned to a strip. A fraction χ of unit electric power $P_s = 1000p_s$ W heats unit supply flow $\dot{m}_s$ before reaching people:

$$h_s^{\mathrm{sup}} = h_{o,s} + \chi P_s/\dot{m}_s, \qquad w_s^{\mathrm{sup}} = w_{o,s}, \qquad 0 \le \chi \le 1. \tag{10}$$

For A, the unheated outlet equals the inlet. Mixing conserves h, w, after which temperature and RH are recovered:

$$w_{\ell,s,z} = (1 - \beta_{s,z})w_b + \beta_{s,z}w_s^{\mathrm{sup}}, \tag{11}$$

$$h_{\ell,s,z} = (1 - \beta_{s,z})h_b + \beta_{s,z}h_s^{\mathrm{sup}}, \tag{12}$$

$$T_{\ell,s,z} = \frac{h_{\ell,s,z} - 2501000w_{\ell,s,z}}{1006 + 1860w_{\ell,s,z}}. \tag{13}$$

RH is not averaged directly. The bus uses $b = a$ and $\chi = 0$. Its B/C fractions imply entrainment-to-supply ratios $(1 - \beta)/\beta = 0.25/1.22/9.00$ and $0.11/0.43/1.50$. Parallel strip branches share one device's total supply.

5.3 A common response and efficiency measure

Standard effective temperature (SET) combines air temperature, radiation, airspeed and humidity for the reference passenger [7, 8]:

$$\Theta_{s,z} = F_{\mathrm{SET}}(T_{\ell,s,z}, T_{r,z}, v_{s,z}, \phi_{\ell,s,z}; \mathrm{met}, \mathrm{clo}). \tag{14}$$

For the assumed hot-discomfort reference $\Theta_c = 26\,^\circ\mathrm{C}$, define

$$D(\Theta) = [\Theta - \Theta_c]_+, \qquad [x]_+ = \max(x, 0). \tag{15}$$

Against the same shelter without active cooling, the signed improvement is

$$b_{s,z,k} = D(\Theta_{0,z,k}) - D(\Theta_{s,z,k}). \tag{16}$$

Adverse effects remain negative. With $n_{z,k}$ people and time steps Δt_k in hours,

$$U_s = \sum_k \sum_z n_{z,k} b_{s,z,k} \Delta t_k, \qquad E_s = \sum_k p_{s,k} \Delta t_k, \tag{17}$$

$$\eta_s = U_s / E_s, \qquad E_s > 0. \tag{18}$$

Here E is kWh and η is person·°C·h/kWh, not COP. Empty zones earn no service; the zero-energy baseline has no ratio. The residual provides an accounting check:

$$R_s = \sum_k \sum_z n_{z,k} D(\Theta_{s,z,k}) \Delta t_k, \qquad \overline{R}_s = \frac{R_s}{\sum_k \sum_z n_{z,k} \Delta t_k}. \tag{19}$$

Thus $U_s + R_s = R_0$. We rank efficiency by η; overall adequacy enters Section 8. Under common operating time, the power ratios give delivery-independent break-even conditions:

$$\begin{aligned} \eta_B > \eta_A &\iff U_B > 4.8 U_A, \\ \eta_C > \eta_A &\iff U_C > 10 U_A, \\ \eta_C > \eta_B &\iff U_C > \frac{25}{12} U_B. \end{aligned} \tag{20}$$

5.4 Results: humidity and spatial delivery

For C, dry-case SET rises from 26.3 °C near the outlet to 29.4 °C at the edge; at RH 70%, it rises from 33.1 to 35.2 °C. These gradients explain why a single air-temperature drop misses delivered service.

Dry-case efficiencies are 84.6/24.5/21.1 for A/B/C; humid-case values are 90.6/21.1/15.2 (Figure 6). B/C outlet drops fall from 5/8 °C to about 3.4/3.4 °C, as their common wet-bulb limit replaces the caps.

Moving the same 12 people toward the edge lowers dry-case A efficiency from 118.3 to 48.3 and reverses B/C ordering: 24.5 > 21.1 under uniform occupancy becomes 15.5 < 16.6 under edge-heavy occupancy. A leads all six nominal climate–occupancy cases.

5.5 Delivery failure and robustness

Aiming or obstruction can weaken delivered airflow without reducing power. Holding $p_A = 0.25$ kW, test

$$v_{A,z}(q) = v_0 + q(v_{A,z}^{\mathrm{nom}} - v_0), \qquad 0 \le q \le 1, \tag{21}$$

q is a delivery factor, not a motor-speed setting. On the sampled grid, A first wins at $q = 0.18$ for RH 30% and 0.16 for RH 70%; the threshold lies within the preceding 0.02 interval. B wins below it.

One-factor endpoint tests give maximum B/A and C/A efficiency ratios 0.432 and 0.372. Including full local electric heating ($\chi = 1$) changes dry-case efficiencies to 81.0/20.8/18.0 and humid values to 86.8/17.4/12.1; A remains first under those inputs. Psychrometric round trips, mass/enthalpy closure, $U + R = R_0$, zero-occupancy and signed-adverse checks pass. The delivery-loss reversal identifies a stronger sensitivity than these individual endpoint perturbations.

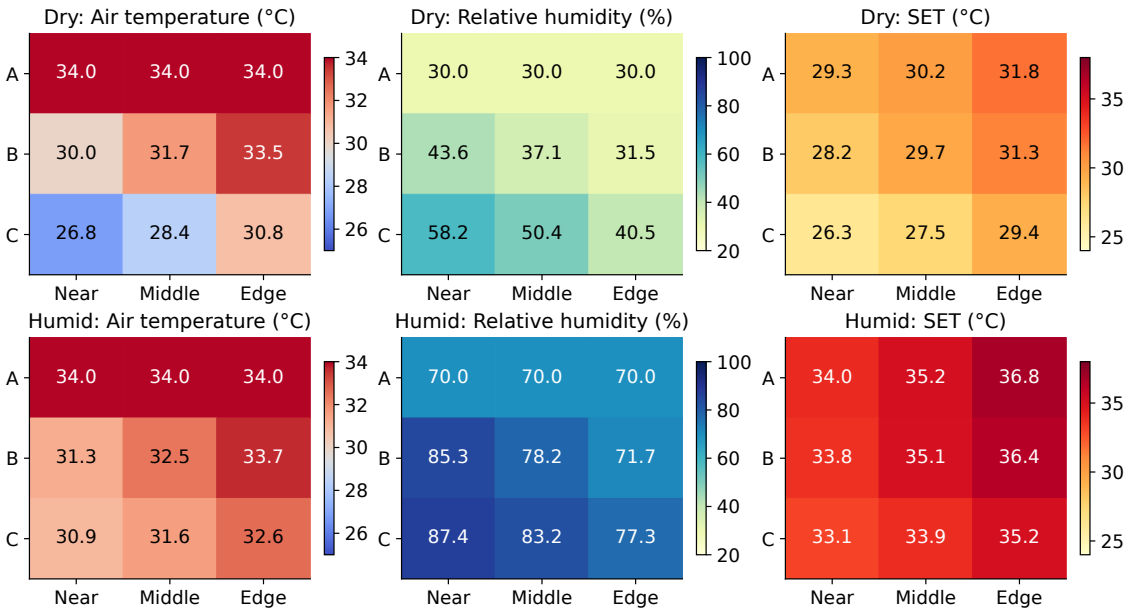


Figure 5: Local conditions under the bus inputs: A changes SET through airflow; B/C also change temperature and moisture.

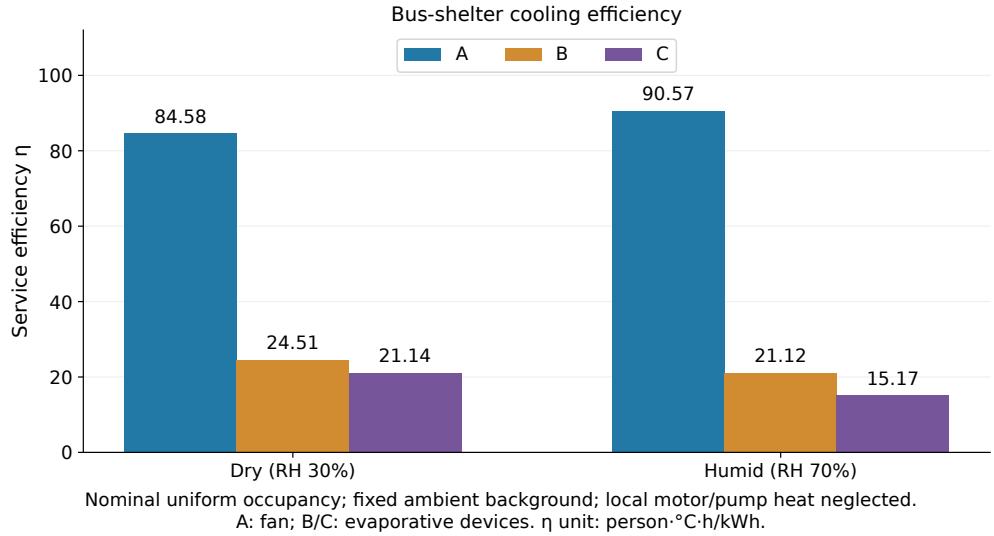


Figure 6: Uniform-occupancy efficiency: A leads both nominal humidity cases.

5.6 Extreme dry heat: testing the efficiency ranking

We test $T_a = T_r = 40\,^{\circ}\mathrm{C}$ at RH 15% and 20%, retaining the device powers, delivery profiles, 5/8 °C outlet caps, and uniform occupancy. Figure 9 shows $\eta_A > \eta_B > \eta_C$ in both cases. At RH 15%, the mean SET reductions are 1.43, 1.95, and 3.42 °C, whereas the efficiencies are 68.43, 19.48, and 16.41 person·°C·h/kWh. C produces greater relief per passenger, but consumes ten times A's electricity. SET relief explains the service numerator; η remains the comparison criterion.

The heat balance explains why dry-bulb cooling alone cannot establish the ranking. When air is warmer than the exposed skin or clothing surface, faster flow increases convective heat gain; it also promotes evaporative heat loss. SET combines these competing exchanges rather than imposing a temperature-triggered reversal of fan performance [7].

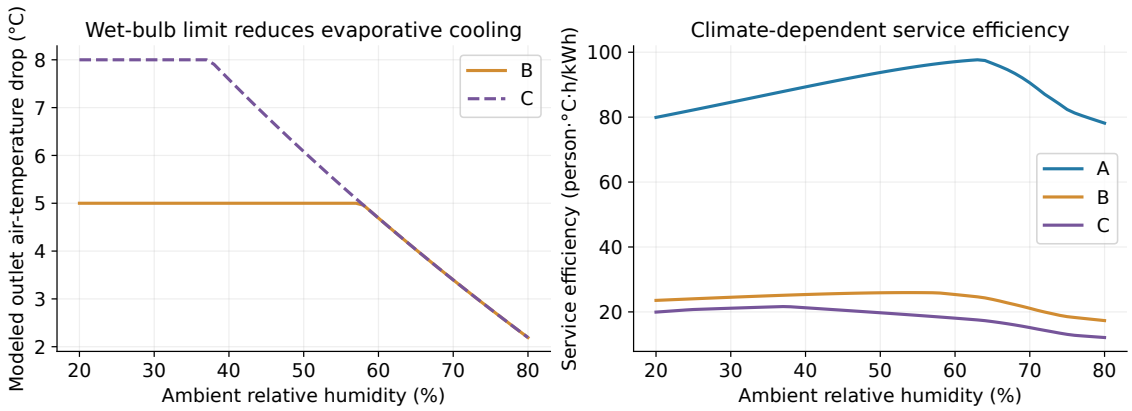


Figure 7: Humidity changes evaporative potential and SET-based efficiency at fixed delivery and demand.

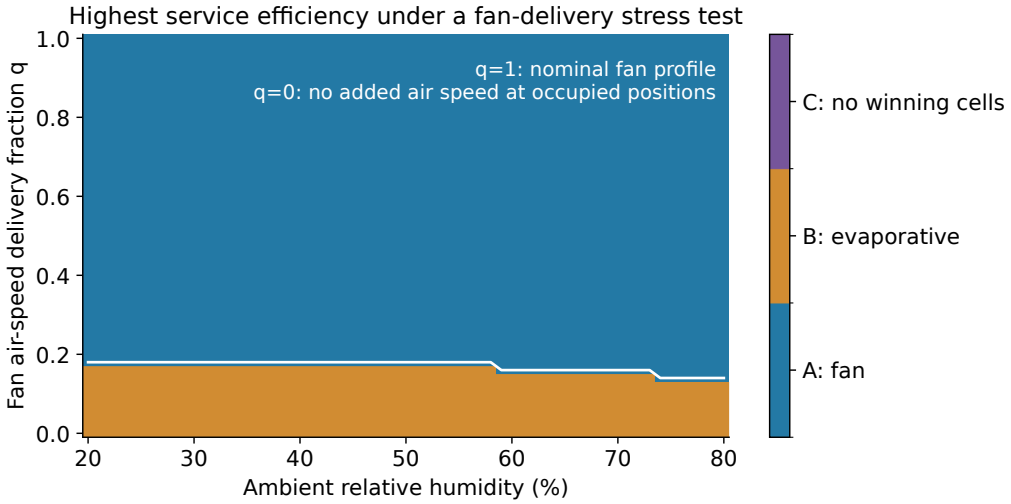


Figure 8: Efficiency winner for RH 20–80% and delivered-speed factor q; C wins no cell of this grid.

6 Canberra Station: Dynamic Background and Local Service

6.1 From a fixed reservoir to coupled zones

Canberra MRT is an elevated, naturally ventilated station in Singapore [15, 16]. Using its approximate $L = 220$ m length with assumed aggregate waiting width $b = 6$ m and usable fraction $\alpha = 0.8$ gives

$$A_{\text{occ}} = \alpha L b = 1056\,\text{m}^2. \tag{22}$$

Three access/core/edge zones each have 352 m^2 and, at 4 m effective height, 1408 m^3. Table 1 defines the 33 °C, RH 70% comparison. Its 4–10 h^{-1} background exchange implies 6–15 min replacement times, comparable to the 15-min passenger cycle: storage and humidity feedback must now be resolved.

Each zone retains h_z, w_z and effective surface temperature $T_{m,z}$, with constant reference dry-air mass $M_z = \rho_{da,o} V_z$. The device, local-mixing and SET operators remain those of the bus. For $x = h$ or w, balanced exchange is

$$\mathcal{L}_z(x) = \dot{m}_{0z}(x_o - x_z) + \sum_{j \neq z} \dot{m}_{zj}(x_j - x_z), \qquad \dot{m}_{0z} = M_z \lambda_z / 3600. \tag{23}$$

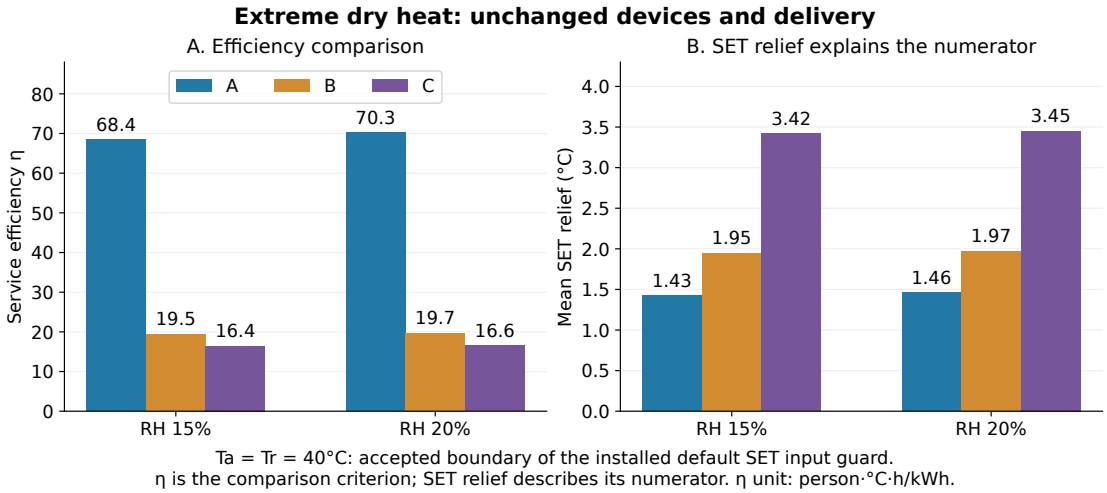


Figure 9: Extreme dry-heat boundary test with the original device powers and delivery profiles. The left panel compares cooling efficiency; the right panel explains its SET-relief numerator. Inputs follow Assumption A4.

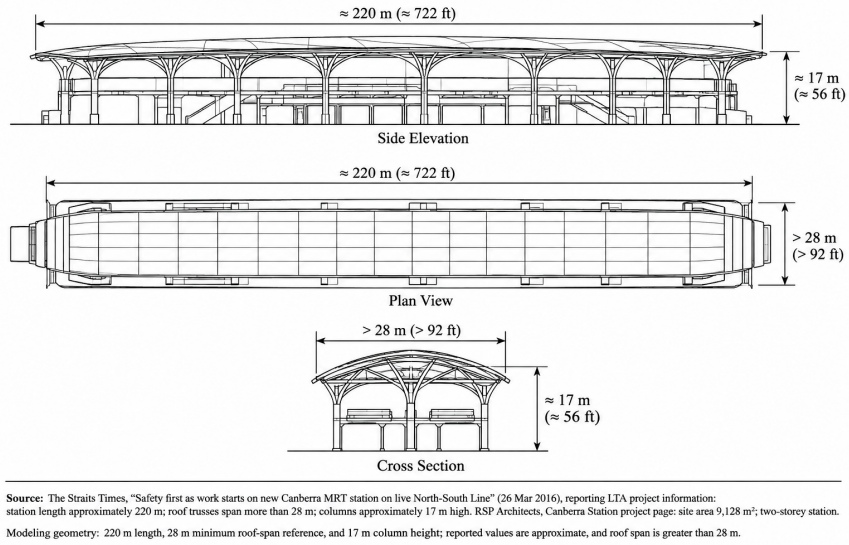


Figure 10: Canberra station context. Structural dimensions guide scale; the three waiting zones are modeled separately [17, 15].

λ_z is ACH in h^{-1}; interzone rates satisfy $\dot{m}_{zj} = \dot{m}_{jz}$. Outdoor exchange represents wind/opening effects through the prescribed cycle [14]. With time in seconds, the three conservation equations per zone are [12]

$$M_z \dot{h}_z = \mathcal{L}_z(h) + H_{ma,z}(T_{m,z} - T_z) + n_z q_c + n_z g h_v(T_z) + P_z + \sum_k x_{zk} \dot{m}_k (h_{e,zk} - h_z), \tag{24}$$

$$M_z \dot{w}_z = \mathcal{L}_z(w) + n_z g + \sum_k x_{zk} \dot{m}_k (w_{e,zk} - w_z), \tag{25}$$

$$C_{m,z} \dot{T}_{m,z} = H_{mo,z}(T_o - T_{m,z}) + Q_{\mathrm{sol},z} + n_z q_r - H_{ma,z}(T_{m,z} - T_z). \tag{26}$$

Here n_z is occupancy; per-person q_c, q_r are convective/radiant W and g is vapor kg/s. Vapor enthalpy is $h_v(T) = 2501000 + 1860T$ J/kg, with no second latent term. H_{ma}, H_{mo} are W/K, C_m is J/K, Q_{sol} is W,

and $P_z = \sum_k x_{zk} P_k$ with $P_k = 1000 p_k$ W. Subscript e denotes the unheated device outlet (the bus's o, s), so A has no evaporative source; B/C add moisture while approximately conserving outlet enthalpy. Surface exchange cancels between the air and wall budgets.

Recover $T_z = (h_z - 2501000 w_z)/(1006 + 1860 w_z)$; effective view fraction F_z determines mean radiant temperature [7]:

$$T_{r,z} = \left[F_z (T_{m,z} + 273.15)^4 + (1 - F_z)(T_{r,o} + 273.15)^4\right]^{1/4} - 273.15. \tag{27}$$

The separate outdoor radiant input is $T_{r,o}$. Increasing w_z reduces the next device pass's wet-bulb depression, providing the feedback absent from the fixed-background bus.

6.2 Local delivery with exact return accounting

Device flow and entrained background return to their original zone. For finite entrainment ($\beta > 0$), the local mixing equations imply

$$(\dot{m}_k + \dot{m}_e)(h_{\ell,zk} - h_z) + (1 - \chi) P_k = \dot{m}_k (h_{e,zk} - h_z) + P_k, \tag{28}$$

$$(\dot{m}_k + \dot{m}_e)(w_{\ell,zk} - w_z) = \dot{m}_k (w_{e,zk} - w_z). \tag{29}$$

Thus the local exposure calculation preserves the existing net heat and moisture sources exactly. Covered people use local T, ϕ, v; uncovered people use background T, ϕ, v_0, and both share T_r. Here $\beta = 0.5$, $\chi = 1$ and A/B/C unit flows are $0.6/0.6/1.2$ kg dry air/s under Table 1; $\chi = 1$ places all electrical heat in supply before passengers.

Two reductions identify the inheritance precisely. At $\beta = 0$, the scalar exposure becomes the original shared-background calculation while prescribed speeds remain. Fixing background to outdoor h_a, w_a, using bus radiation, speeds, occupancy and β, and setting $\chi = 0$ recovers the bus calculation exactly. Setting time derivatives to zero alone would not impose that reservoir boundary.

6.3 Counts, coverage and efficiency

Let s index an installation and x_{zk} its nonnegative integer count of type k in zone z. For unit footprints $a_{A/B/C} = 20/30/40\,\mathrm{m}^2$, uniform zone density gives

$$c_{zk} = \frac{x_{zk} a_k}{A_z}, \qquad c_{z0} = 1 - \sum_k c_{zk} \geq 0, \qquad \sum_k x_{zk} a_k \leq A_z. \tag{30}$$

These passenger fractions c differ from the air-mixing fraction β. Average discomfort across subgroups, not velocities or device efficiencies:

$$\overline{D}_{s,z}(t) = c_{z0} D[\Theta(T_z, T_{r,z}, v_0, \phi_z)] + \sum_k c_{zk} D[\Theta(T_{\ell,zk}, T_{r,z}, v_{zk}, \phi_{\ell,zk})]. \tag{31}$$

All units affect the background; direct coverage determines the additional local exposure. Every plan starts after the same two-hour uncooled warm-up, runs two hours, and uses a matched no-device baseline. The common passenger history totals 440 person-hours. With p_k in kW,

$$U_s = \frac{1}{3600} \int_0^{7200} \sum_z n_z(t) [D_{0,z}(t) - \overline{D}_{s,z}(t)]\, dt, \tag{32}$$

$$E_s = \frac{1}{3600} \int_0^{7200} \sum_{z,k} x_{zk} p_k\, dt, \qquad \eta_s = U_s / E_s. \tag{33}$$

The $1/3600$ factors convert seconds to hours, retaining person·°C·h/kWh. This section compares U/E. Mean residual $\overline{R}_s = \int \sum_z n_z \overline{D}_{s,z}\, dt / \int \sum_z n_z\, dt$ supplies the later adequacy criterion in Section 8.

6.4 Deployment results and verification

Among 24 sampled plans, A3 maximizes η at 37.58; among the six equal-68.18%-coverage benchmarks, A36 leads at 35.49. The distinct candidate sets answer scale and device-comparison questions, respectively (Figure 11).

Plan suffixes give total device counts. For A36/B24/C18/AB30/AC27/ABC24, per-zone (A, B, C) counts are $(12, 0, 0)$, $(0, 8, 0)$, $(0, 0, 6)$, $(6, 4, 0)$, $(6, 0, 3)$ and $(3, 2, 3)$, respectively; A3 places one fan in each zone.

Table 4: Nominal efficiency accounting: U in person·°C·h, E in kWh.

Plan	Person-time coverage	U	E	η
A3	5.68%	56.38	1.5	37.58
A36	68.18%	638.89	18.0	35.49
B24	68.18%	446.28	57.6	7.75
C18	68.18%	453.00	90.0	5.03
AB30	68.18%	581.61	37.8	15.39
AC27	68.18%	593.68	54.0	10.99
ABC24	68.18%	524.76	63.9	8.21
AB30-R	76.19%	695.07	37.8	18.39

Redistributing AB30's 18 A and 12 B units to zone counts $(3, 4, 0)$, $(12, 3, 0)$ and $(3, 5, 0)$ raises person-time coverage from 68.18% to 76.19%. At the same 37.8 kWh, U rises from 581.61 to 695.07 and η from 15.39 to 18.39: allocation changes useful delivery without adding equipment.

A36 retains its equal-coverage efficiency lead for all 15 combinations $\beta = 0/0.25/0.5/0.75/1$, $\chi = 0/0.5/1$. Varying A's unit flow through $0.3/0.6/1.2$ kg dry air/s gives efficiencies $34.06/35.49/36.22$. Figure 12 shows the controlled delivery sensitivity.

Global energy/moisture budgets and local return identities close numerically. Halving the time step changes the tested mixed-plan efficiencies by at most 5.4×10^{-6} relatively; $\beta = 0$ reproduces all 24 former shared-background results exactly. The fixed-reservoir test recovers bus states and service. These checks verify the extension's conservation and reduction properties.

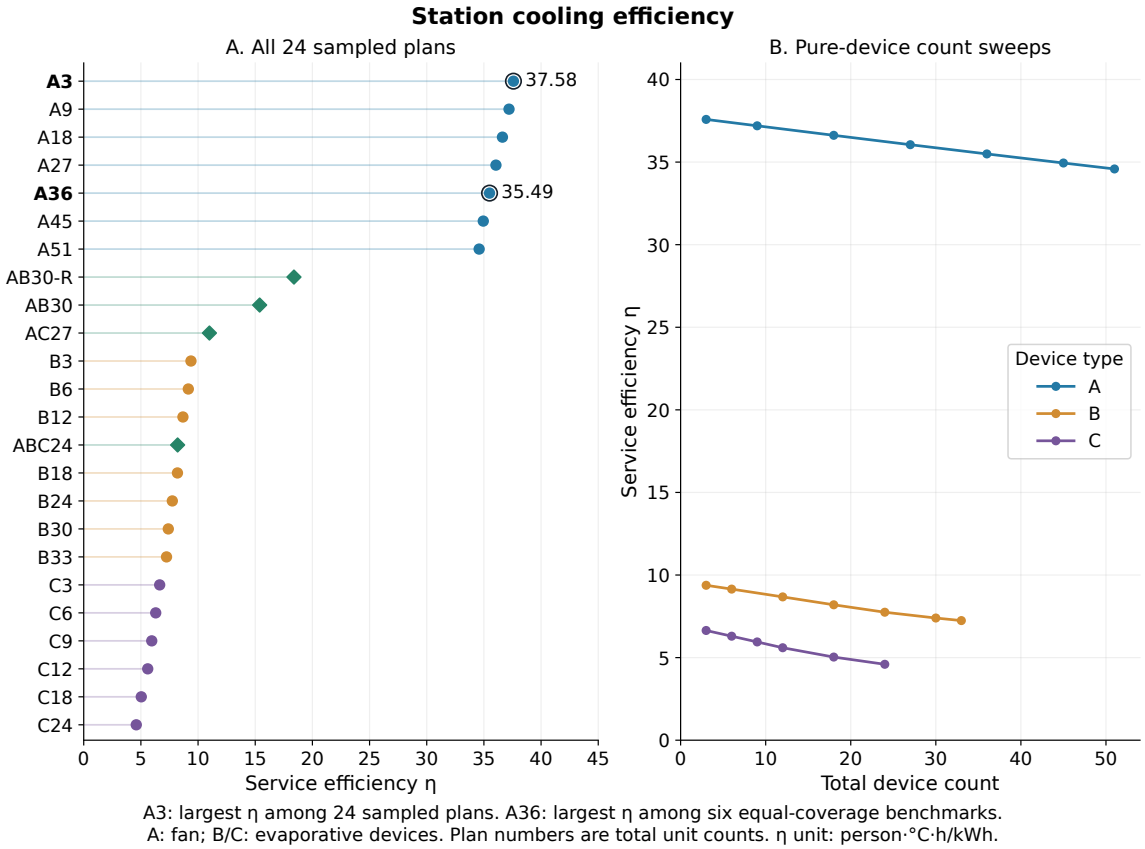


Figure 11: A3 leads all 24 sampled installations; A36 leads the six equal-coverage benchmarks. Device counts change both delivered service and background loads.

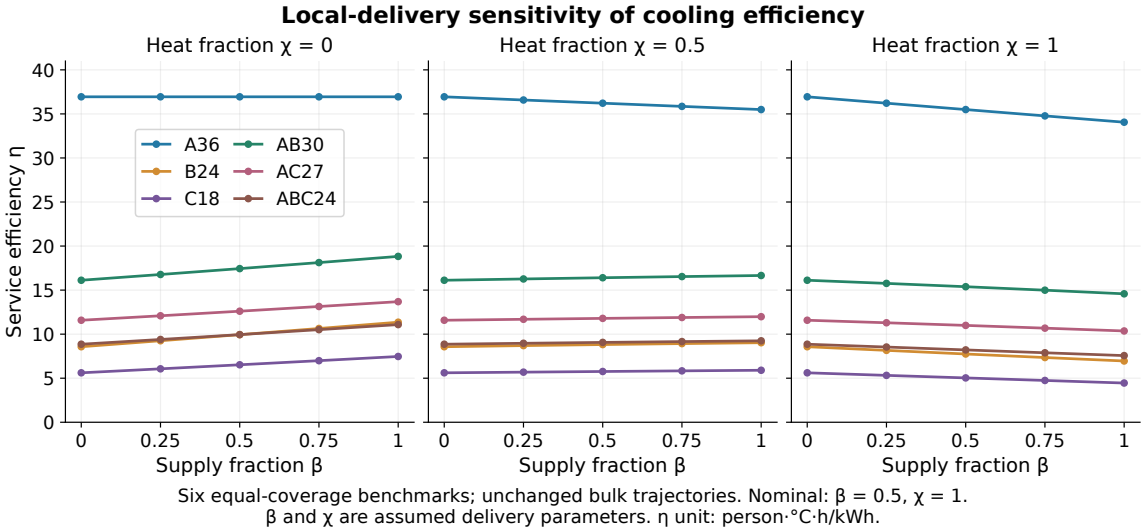


Figure 12: Efficiency sensitivity to mixing and heat allocation; background trajectories remain unchanged by the local mapping.

7 Apply the Model: From a Ticket Queue to Fremont Street

7.1 Two service areas, a 140-fold difference in size

Fremont Street Experience, Las Vegas, provides a large covered pedestrian setting: its official canopy length is 1,375 ft, or 419.1 m [19]. Our three demand zones span an 8382-m^2 occupied strip. At the **Honolulu Zoo entrance**, the architect explicitly designed shaded ticket-purchasing and queuing space [20]; the modeled service boundary is the 60-m^2 public queue outside the ticket window. The area ratio is 139.7. Las Vegas supplies the dry-desert case; Honolulu supplies a warm coastal case with a separate natural-wind sensitivity [22, 23]. Geometry, attendance and weather inputs are collected in Table 3.

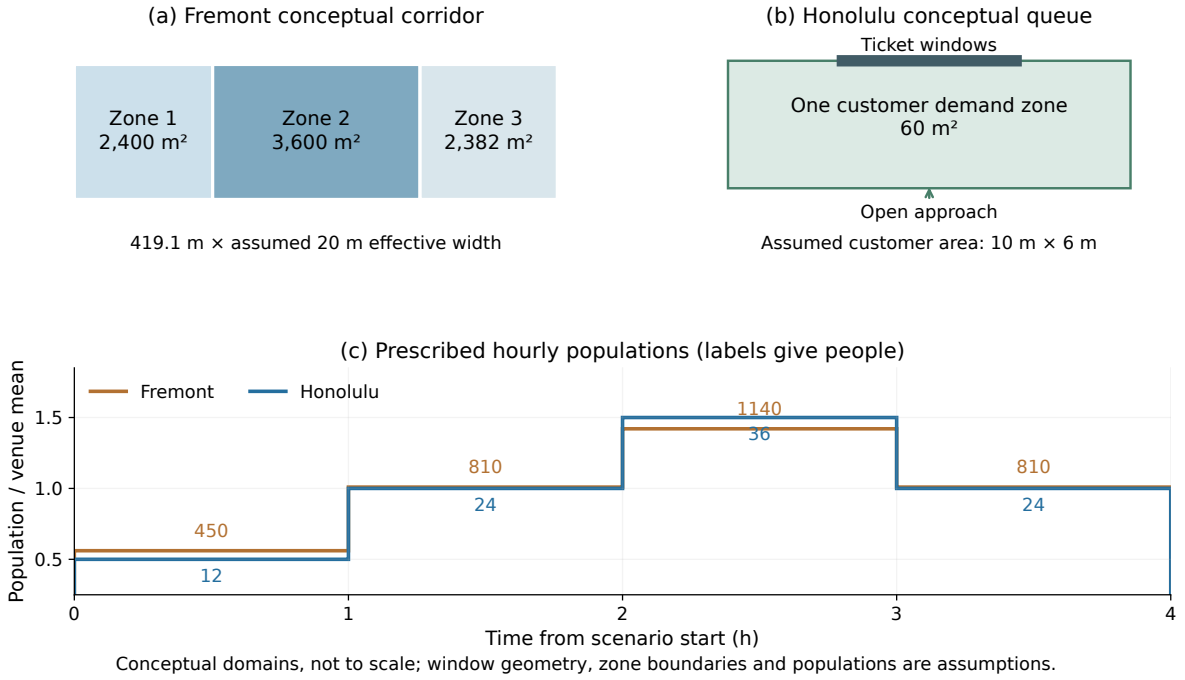


Figure 13: Analytical service boundaries and prescribed attendance. Fremont has three demand zones; the ticket queue has one. The three local delivery strips belong to each device footprint, independently of this site partition.

7.2 Transfer the physical kernel, then aggregate service

We use the fixed-reservoir limit of the station model: outdoor h, w, T_r are prescribed, while its integer allocation and coverage calculation are retained. Each unit uses the same wet-bulb-limited outlet, conservative local mixing and SET response as in Task 2. The three strip responses are averaged *after* conversion to excess SET. Let $\delta_k = \frac{1}{3}\sum_{j=1}^{3}[D_0 - D(\Theta_{kj})]$ be the mean relief inside a complete footprint, and $\mathcal{H}_z = \int_0^H n_z(t)\,dt$ the zone's visitor-hours. Then

$$u_{zk} = \mathcal{H}_z \frac{a_k}{A_z}\delta_k, \qquad U_s = \sum_{z,k} x_{zk}u_{zk}, \qquad E_s = H\sum_{z,k} x_{zk}p_k. \tag{34}$$

Time is measured in hours, giving 3210 and 96 visitor-hours in Fremont and Honolulu. Person-time coverage is $q_s = \sum_z \mathcal{H}_z \sum_k c_{zk} / \sum_z \mathcal{H}_z$. All installed units run throughout the four-hour period; water

supply is counted once per unit. The disjoint-footprint condition in Eq. (30) prevents double-counting people.

This linear service structure also solves the efficiency-only allocation problem. For every positive-energy layout,

$$\eta_s = \sum_{z,k} \frac{Hx_{zk}p_k}{E_s} \frac{u_{zk}}{Hp_k} \le \max_{z,k} \frac{u_{zk}}{Hp_k}. \tag{35}$$

The bound is attained by one unit of the maximizing type in the maximizing zone whenever its footprint fits. It yields 16.54 **in Fremont and** 47.71 **in Honolulu** person·°C·h/kWh, attained by A in Fremont's middle demand zone and A in the ticket queue. Their coverages are 0.33% and 33.33%. Additional A units within the same zone tie while their footprints fit. This efficiency result uses the entire integer domain, without a service-coverage or cost threshold.

7.3 Climate changes the relative value of the devices

Table 5 compares the largest pure-type installations that fit each site. Honolulu admits three A units, two B units or one C unit: the last serves only two-thirds of the queue. Every unit's full power remains in the denominator.

Table 5: Largest-fitting pure installations. Coverage q is visitor-time weighted; U has units person·°C·h and $\eta = U/E$.

Venue	Plan	q (%)	U	E (kWh)	η
Fremont	A419	99.99	4977.81	419.0	11.88
	B279	99.91	6782.04	1339.2	5.06
	C209	99.84	11951.71	2090.0	5.72
Honolulu	A3	100.00	143.12	3.0	47.71
	B2	100.00	204.26	9.6	21.28
	C1	66.67	206.30	10.0	20.63

The wet-bulb temperatures are 20.77°C and 25.56°C. C's outlet drop reaches 8.00°C in Fremont and 3.81°C in Honolulu. At the dry site, C's larger temperature reduction and delivery profile outweigh its extra power relative to B; at the coastal site, B supplies more useful service per kWh than C. A retains the highest efficiency in both nominal cases. These rankings follow from local exposure and electricity together.

Honolulu's larger efficiency values also reflect its greater visitor density. To isolate the climate effect, we exchange $(T_a, \mathrm{RH}, T_r, v_0)$ between sites while holding geometry, attendance and device counts fixed. The B–C order follows the weather tuple in both transfers. At Honolulu's density, A's efficiency changes from 47.71 under its own weather to 49.63 under Fremont's weather. Thus a high site-level η cannot be attributed to favorable climate alone.

Natural wind also changes the marginal value of equipment: raising Fremont's background speed from .20 to .80 m/s changes the efficiency-maximizing type from A to C, with a new bound of 5.38. The uncooled baseline SET falls from 33.77 to 31.51°C. Extra fan speed brings less benefit when ambient air is already moving, whereas evaporative temperature reduction remains available.

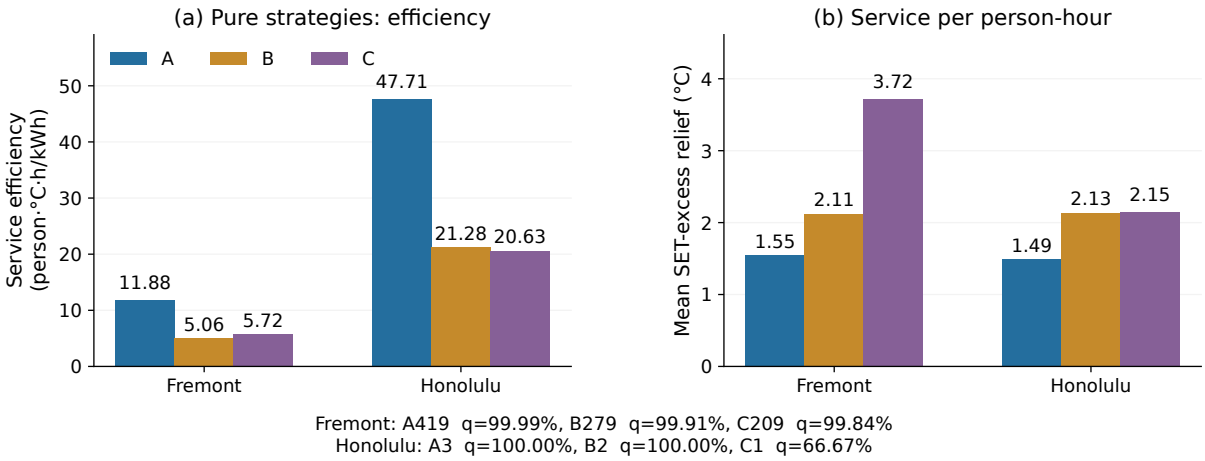


Figure 14: Pure-type installations: useful service per kWh and mean SET-excess relief across all visitors. Coverage differences, including Honolulu C1's uncovered third, remain in the service calculation.

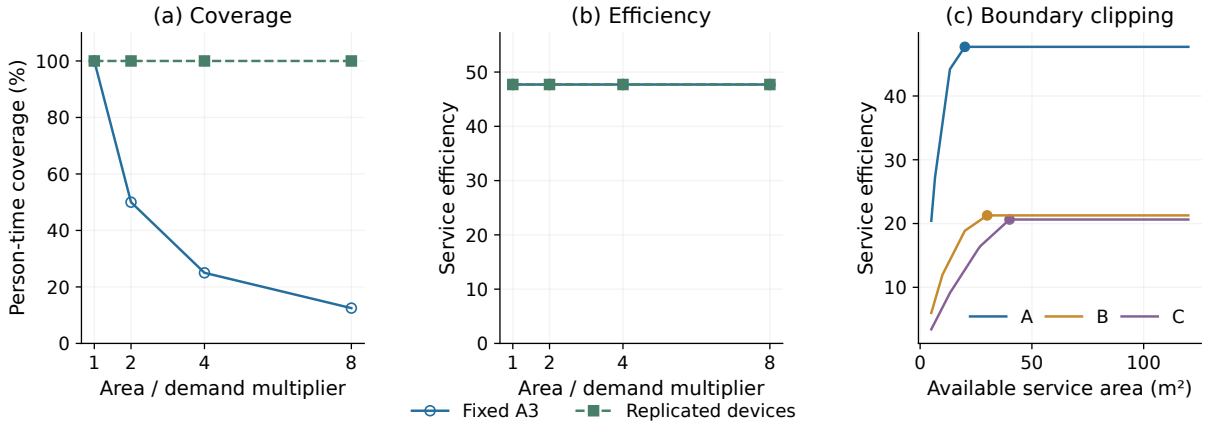
7.4 Controlled scaling and the finite-footprint limit

For a fixed climate and unchanged local delivery, multiply each zone's area and visitor-hours by λ. With the original devices, each footprint contains the same number of people, so U and E remain constant while person-time coverage becomes q/λ. Replicating devices as well gives

$$(A_z, \mathcal{H}_z, x_{zk}) \mapsto (\lambda A_z, \lambda \mathcal{H}_z, \lambda x_{zk}) \quad \Longrightarrow \quad (U, E, q, \eta) \mapsto (\lambda U, \lambda E, q, \eta). \tag{36}$$

Integer multipliers preserve feasible counts. For the Honolulu A3 reference, scaling from 60 to 480 m^2 with attendance at the same density leaves $\eta = 47.71$. Three fixed fans reduce coverage from 100% to 12.5%; 24 fans restore full coverage and raise four-hour electricity from 3 to 24 kWh. These are controlled similarity properties of the model, separating scale from climate.

Small spaces also expose a finite-footprint limit. We retain only the portions of the near, middle and far strips lying within the service boundary and recompute their area-weighted relief. The efficiency plateaus once a complete footprint fits; below that size, lost served area changes the result even at unchanged visitor density. Therefore the transferable characteristics are **occupied density, retained local delivery, wet-bulb depression and footprint fit**. Geometry sets who benefits; climate sets the benefit per exposed person. Matched-baseline identities, valid local air states and the scaling relations are checked against the exported calculations.



Honolulu climate and fixed density. Efficiency unit: person·°C·h/kWh.
Clipping removes far, then middle strips; full power is charged. Markers: complete device footprint.

Figure 15: Controlled scale and boundary tests. Proportional deployment preserves efficiency and coverage; fixed equipment preserves efficiency but loses coverage. In the separate single-unit test, reducing available area clips the far delivery strip first, with the full unit power still charged.

8 Overall Effectiveness: Four Objectives, an Explicit Choice

8.1 Performance, access, energy and annual cost

Task 3 ranks useful service per kWh. Task 4 adds the operator's service and resource requirements. We compare

$$\min_s \left(\overline{R}_s,\ X_s,\ E_{s,y},\ C_{s,y}\right), \qquad X_s = \frac{\sum_{z,k} x_{zk} b_k}{\sum_z A_z}. \tag{37}$$

Here $\overline{R}_s$ is remaining SET excess; b_k is equipment and maintenance clearance in m^2; and X_s is the fraction of public service area lost, a *visitor-access proxy*. Clear routes and exits remain layout requirements. Annual electricity $E_{s,y}$ and expenditure $C_{s,y}$ describe resource use and cost. Thermal relief is counted once; η is not a fifth objective.

Using Table 3, let $E_{s,y} = H_y \sum_{z,k} x_{zk} p_k$ and $\mathrm{CRF} = r(1+r)^L/[(1+r)^L - 1]$. Then

$$\begin{aligned} C_{s,y} &= \mathrm{CRF} \sum_{z,k} x_{zk} K_k + \sum_{z,k} x_{zk} M_k + \pi_e E_{s,y} + \pi_w V_{s,y}, \\ V_{s,y} &= \frac{3600 H_y \lambda_w}{\rho_w} \sum_{z,k} x_{zk} \dot{m}_k (w_{e,k} - w_a). \end{aligned} \tag{38}$$

K_k is capital cost (USD), M_k annual maintenance (USD/year), H_y hours/year and ρ_w water density (kg/m^3). Thus $E_{s,y}$, $V_{s,y}$ and $C_{s,y}$ have units kWh/year, m^3/year and USD/year. Water is charged once per unit, including the make-up factor λ_w.

8.2 An integer minimum-cost service policy

An ε-constraint policy minimizes cost while setting explicit limits on the other outcomes:

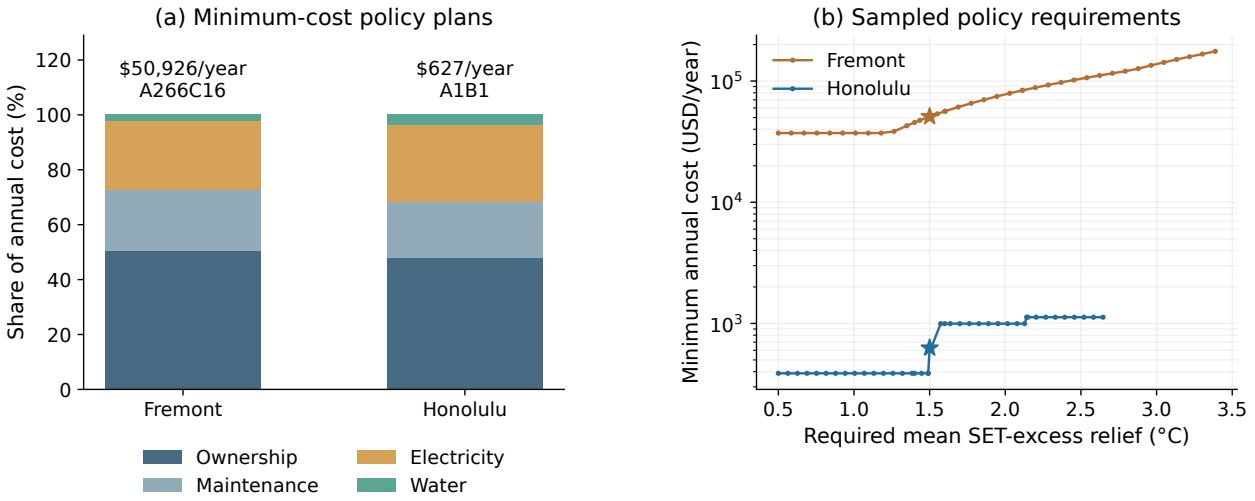
$$\begin{aligned} s^* = \arg\min_{s} C_{s,y}, \qquad & x_{zk} \in \mathbb{Z}_{\geq 0}, \\ \sum_{k} a_k x_{zk} \leq A_z \quad & (\text{every } z), \qquad q_s \geq 0.80, \\ \frac{U_s}{\sum_z \mathcal{H}_z} \geq 1.5^{\circ}\text{C}, \qquad & X_s \leq 0.035, \qquad E_{s,y} \leq e_{\max} \sum_{z} A_z. \end{aligned} \tag{39}$$

Here a_k is the service footprint and $e_{\max} = 30$ kWh/(m^2·year). The mean-relief constraint is equivalent to a residual ceiling under each venue's fixed baseline. All coefficients are fixed by the physical model, giving nine integer variables in Fremont and three in Honolulu. The mixed-integer linear solver reports zero optimality gap for both nominal cases. This establishes the minimum cost over all feasible integer allocations under the stated policy.

Table 6: Minimum annual cost under Eq. (39). Relief includes uncovered visitors; annual energy and cost are in kWh/year and USD/year.

Venue	Plan	q (%)	Relief (°C)	X (%)	E_y	C_y
Fremont	A266C16	81.79	1.5002	1.87	63,900	50,926
Honolulu	A1B1	83.33	1.5608	2.50	870	627

One optimal Fremont assignment is (116A, 148A + 16C, 2A) across the three zones. It directs service toward high visitor density; the third zone receives only 1.68% coverage. A requirement to serve every sector would need additional zonal coverage constraints. The selected plans leave mean SET excesses of 6.274°C and 3.902°C, respectively, so meeting the relief target does not imply thermal neutrality.



Coverage ≥80%; floor clearance ≤3.5%; annual energy ≤30 kWh/m². Stars: 1.5°C requirement.

Figure 16: Annual cost composition and minimum cost versus required mean relief, with the other policy limits fixed. Stars mark the 1.5°C policy; the cost axis in the right panel is logarithmic. Integer choices create discrete transitions.

8.3 Threshold sensitivity changes the recommendation

In Honolulu, A3 provides full coverage and 1.4908°C relief: only 0.0092°C below the nominal target. Targets of 1.40 or 1.49°C select A3 at USD 388/year; 1.50°C selects A1B1 at USD 627/year; 1.60°C selects B2 at USD 996/year. Thus the thermal service requirement controls the preferred configuration.

Across 243 price combinations, Honolulu retains A1B1 throughout; Fremont retains its nominal counts in 56 cases, while 69 select pure A398. Physical changes also matter: increasing Honolulu's background speed from 0.35 to 0.8 m/s selects A1C1 because the required benefit is measured above the improved natural-wind baseline. Reducing its queue area to 45 m^2 makes the unchanged policy infeasible under full-footprint packing. Operators should establish service targets, usable area and delivery conditions before committing to a configuration.

Memorandum to the Chief Operations Officer

To: Chief Operations Officer, HiMCM
From: Cooling-service modeling team
Subject: A service-targeted cooling plan for the Fremont canopy corridor

For the modeled 8382-m^2 occupied corridor beneath the Fremont Street Experience canopy [19], use **266 type-A fans and 16 type-C evaporative units** as the nominal planning configuration. At 38°C and 20% RH, this is the minimum annual-cost integer plan meeting our global coverage, mean-relief, clearance and electricity limits. The four-hour attendance profile represents 3210 visitor-hours, with the largest crowd in the middle demand zone.

One optimal assignment puts 116 fans in zone 1, 148 fans and all 16 evaporative units in zone 2, and two fans in zone 3. It provides **81.79% person-time coverage** and 1.5002°C mean reduction in SET excess. Zone coverages are 96.7%, 100% and 1.7%: the global objective strongly favors the busiest sector. If service is required in every sector, add zonal coverage constraints before procurement. Equipment clearance occupies 1.87% of the total service area; circulation routes and exits still require a layout check.

A four-hour operating period uses **426 kWh and approximately 2540 liters of make-up water**, delivering 4816 person·°C·h of relief. Efficiency is 11.30 such service units per kWh. Mean residual SET excess remains 6.27°C, so this configuration reduces heat burden without making the corridor thermally neutral.

The common price basis gives **USD 111,800 in equipment capital** and **USD 50,926/year** at 600 operating hours. The annual total comprises approximately USD 25,823 capital recovery, 11,180 maintenance, 12,780 electricity and 1,143 water. Obtain installation and utility quotes before purchase: the nominal counts persist in only 56 of 243 tested price combinations, and 69 combinations favor fans alone. A 1.4°C relief target instead selects 280 fans and nine C units at USD 45,502/year; a 1.6°C target selects 259 fans and 22 C units at USD 56,221/year.

Commission a representative section before full deployment. Measure air speed, temperature and humidity near, midway from and at the edge of each unit; confirm delivery to standing visitors, water supply and unobstructed movement. Update the allocation using the observed crowd distribution and conditions. The nominal mean-relief margin is very small, so the computed optimum offers no delivery reserve. Agree a margin or revised service target before implementation, and reoptimize when weather or attendance changes.

During severe heat, follow National Weather Service guidance: monitor forecasts, adjust strenuous outdoor activity, and provide breaks and access to cooler locations [24]. Use official heat advisories to trigger measures beyond equipment operation.

Public Blog: How Much Comfort Does One Kilowatt-Hour Buy?

An outdoor fan can help people feel cooler while the air thermometer barely changes. An evaporative cooler lowers air temperature while adding moisture. A fair comparison follows the cooling to the visitor: what temperature, humidity, air movement and radiant heat reach that person, and for how long?

Our model combines those influences using Standard Effective Temperature, or SET. One service unit means reducing one person's SET excess by one degree for one hour. Dividing that benefit by electricity gives cooling efficiency. It measures relief accumulated over visitor-hours.

We applied the same calculation to two very different spaces: a large occupied corridor under the Fremont Street Experience canopy in Las Vegas, and a shaded ticket queue at Honolulu Zoo [19, 20]. The modeled areas are 8382 and 60 square meters. Las Vegas offers dry heat; Honolulu offers a warm, humid coastal setting. The smaller queue also packs more visitor-time into each square meter, which helps each unit serve more people.

At Honolulu, three fans cover the queue and deliver about **143 service units using 3 kWh**: roughly **48 units per kWh**. In Fremont, placing one fan in the busiest demand zone yields about **17 units per kWh**, but directly serves only **0.33% of visitor-time** across the corridor. A good efficiency ratio can coexist with very limited coverage. More equipment in the same busy zone can retain that ratio while serving more people.

Choosing a practical installation adds another question: how much service must it provide, and at what cost? Our illustrative policy requires at least 80% coverage and 1.5°C average relief, alongside space and electricity limits. It selects 266 fans plus 16 evaporative units in Fremont, and one fan plus one evaporative unit in Honolulu. In the queue, three fans already deliver 1.4908°C relief. Lowering the target from 1.50 to 1.49°C changes the least-cost answer to fans alone.

Scale brings a similar distinction. If Honolulu's area and crowd both grow eightfold, keeping three fans preserves their efficiency but lowers coverage from 100% to 12.5%. Twenty-four fans restore coverage. The useful questions are therefore connected: how much benefit does each kWh buy, who receives it, and what service can the operator afford? During severe heat, also follow local advisories and adjust outdoor plans [24].

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Report on Use of AI Tools

OpenAI Codex was used during November 2026 to assist with translation, model formulation, Python implementation, source research, English drafting, LaTeX preparation and figure generation. The user also supplied a Google Gemini analysis as a PDF excerpt; its process-oriented explanation informed the early organization. Gemini was not used as a scientific source for the physical equations or numerical inputs.

Development record. The assistance covered freezer service accounting; psychrometric and SET calculations; the shared local-delivery operator; station heat, moisture and surface balances; candidate deployment comparisons; scope separation between efficiency and effectiveness; and the final CP-SET restructuring. The supplied drawings and earlier award papers were used for context and presentation, with their data provenance distinguished from scenario inputs.

Latest revision. The user selected Fremont Street Experience and the Honolulu Zoo ticketing queue for a large–small venue comparison. Codex researched site identity and climate, specified the assumption table, implemented the shared CP-SET calculation, and generated climate-transfer, scale and boundary tests. The revision uses an analytical efficiency bound and integer cost optimization, with a separate enumeration audit for the small venue. Associated summaries, operating guidance and standalone PDF figures were updated from the computed results. The Python implementation, input files, CSV outputs, figure scripts and verification records are retained with the LaTeX project.

Verification record. Automated checks cover heat and water conservation, service-energy identities, limiting-case recovery, time-step convergence and consistency between tables, figures and outputs. The AI-assisted workflow performed these checks; independent team review is recorded separately by the team.

Tool identification and transcript archive. OpenAI Codex, accessed November 8, 2026; the session identifies Codex as based on GPT-6; a more specific deployment identifier was not supplied. Google Gemini, user-supplied conversation excerpt; the model identifier is not present in that excerpt. The original prompts and complete retained outputs are held in the associated conversation records. This narrative is an activity record, not a replacement for those transcripts.

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